Options :

6406534214572. ✖ Territorial

6406534214573. ✖ Reputable

6406534214574. ✖ Satisfied

6406534214575. ✔ Both territorial and reputable

Question Number : 198 Question Id : 6406531250866 Question Type : MCQ
Correct Marks : 1
Question Label : Multiple Choice Question
"I will give you the bedroom next to the sitting room."

This sentence is in the _____ aspect.
Options :

6406534214576. ✖ Future perfect

6406534214577. ✖ Future perfect continuous

6406534214578. ✔ Future indefinite

6406534214579. ✖ Simple present

Question Number : 199 Question Id : 6406531250867 Question Type : MCQ
Correct Marks : 1
Question Label : Multiple Choice Question
Choose the correct modal verb to fill in the blank.

____ *you mind if I stayed back after class for remedial classes?*
Options :

6406534214580. ✔ Would

6406534214581. ✖ Could

Sw Testing

Section Id :	64065387595
Section Number :	7
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	19
Number of Questions to be attempted :	19
Section Marks :	100
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No

Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653194242
Question Shuffling Allowed :	No

Question Number : 200 Question Id : 6406531250868 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : SOFTWARE TESTING (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406534214582. ✔ YES

6406534214583. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653194243
Question Shuffling Allowed :	Yes

Question Number : 201 Question Id : 6406531250869 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A user fills out a web form and clicks the “Submit” button. This action causes the application to transition from the form page to another page (e.g., a confirmation or result page). In the context of an Application Transition Graph (ATG), what type of transition does this represent?

Options :

6406534214584. ✖ Simple Link Transition

6406534214585. ✖ Operational Transition

6406534214586. ✔ Form Link Transition

6406534214587. ✖ Redirect Transition

Question Number : 202 Question Id : 6406531250870 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

An online booking system requires users to provide a “check-in date” and a “check-out date” while reserving a hotel room. A tester alters the “check-out date” in the form submission to a date earlier than the “check-in date” to check if the application properly enforces the relationship between the two fields. What type of testing does this scenario describe?

Options :

6406534214588. ✖ Control flow level bypass testing

6406534214589. ✔ Parameter level bypass testing

6406534214590. ✖ Value level bypass testing

6406534214591. ✖ User session data based testing

Question Number : 203 Question Id : 6406531250871 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Which type of testing is performed by suddenly increasing the number of users to a very large amount in order to measure the system’s performance and determine if it can sustain the workload?

Options :

6406534214592. ✔ Spike testing

6406534214593. ✖ Endurance (soak) testing

6406534214594. ✖ Compatibility testing

6406534214595. ✖ Regulatory testing

Question Number : 204 Question Id : 6406531250872 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Which of the following metrics is categorized as a Maintenance Quality Metric?

Options :

6406534214596. ✖ Defect Density

6406534214597. ✔ Backlog Management Index (BMI)

6406534214598. ✖ Code Coverage

6406534214599. ✖ Cyclomatic Complexity

Question Number : 205 Question Id : 6406531250873 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the following method: double root(int a, int b, int c).

```
public double root(int a, int b, int c) {
    return (-b + Math.sqrt(b*b - 4*a*c)) / (2*a);    //LINE-1
}
```

A mutation operator is applied to LINE-1 to get the following mutant:

```
public double root(int a, int b, int c) {
    return (-b + Math.sqrt(b*b - 4*a*b)) / (2*a);    //LINE-1
}
```

Identify the type of the mutation operator.

Options :

6406534214600. ✔ Scalar variable replacement

6406534214601. ✖ Unary operator insertion

6406534214602. ✖ Arithmetic operator replacement

6406534214603. ✖ Absolute value insertion

Question Number : 206 Question Id : 6406531250875 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Why is Test-Driven Development (TDD) ideal for pair programming?

Options :

6406534214608. ✖ Because it eliminates the need for collaboration between developers.

6406534214609. ✔ Because it allows one developer to write code and the other to test.

6406534214610. ✖ Because it requires no test frameworks.

6406534214611. ✖ Because TDD does not involve testing during development.

Question Number : 207 Question Id : 6406531250879 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

In a ride-sharing application, the User Module sends a ride request to the Driver Module via a central messaging service. The modules communicate using message queues. What type of coupling is used in this scenario?

Options :

6406534214624. ✖ Parameter coupling

6406534214625. ✖ Shared data coupling

6406534214626. ✖ External device coupling

6406534214627. ✔ Message-passing interfaces

Sub-Section Number :	3
Sub-Section Id :	640653194244
Question Shuffling Allowed :	Yes

Question Number : 208 Question Id : 6406531250874 Question Type : MCQ
Correct Marks : 5
Question Label : Multiple Choice Question

Given a context free grammar over a finite alphabet $\sum = \{a, b\}$, with the production rules as follows:

$$S \rightarrow aSbb,$$
$$S \rightarrow abb.$$

Let S be the starting variable. Which of the following sets corresponds to the language generated by the given grammar?

Options :

6406534214604. ✔ $\{a^n b^{2n} \mid n \geq 1\}$

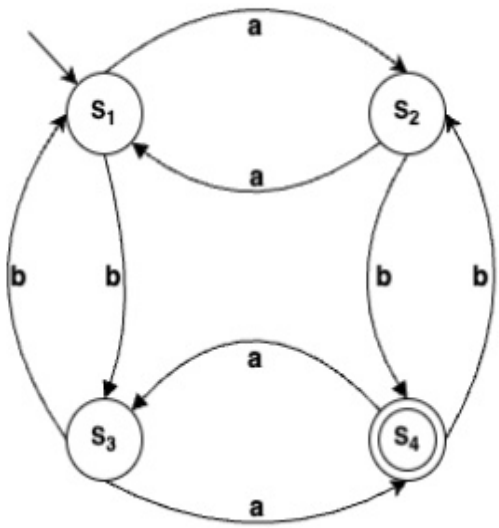
6406534214605. ✖ $\{a^{2n} b^n \mid n \geq 1\}$

6406534214606. ✖ $\{a^n b^n \mid n \geq 0\}$

6406534214607. ✖ $\{a^n b^{2n} \mid n \geq 0\}$

Question Number : 209 Question Id : 6406531250876 Question Type : MCQ
Correct Marks : 5
Question Label : Multiple Choice Question

Consider the below finite state machine (FSM) and choose the correct option.



Options :

6406534214612. ✖ The FSM accepts any string of even numbers of a's and odd number of b's.

6406534214613. ✖ The FSM accepts any string of odd numbers of a's and even number of b's.

6406534214614. ✔ The FSM accepts any string of odd numbers of a's and odd number of b's.

6406534214615. ✖ The FSM accepts any string of even numbers of a's and even number of b's.

Question Number : 210 Question Id : 6406531250877 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Consider the code given below.

```
int compute(int x, int y, int z){
    boolean r = false;
    if (x > y)
        r = true;

    if(r && x == z)
        return x;
    else
        return z;
}
```

Which of the following lists test cases for RACC for the function `compute`?
Please note that the variable `r` needs to be re-written in terms of inputs in the second predicate to achieve various coverage criteria. We use the predicate `x > y` to represent `r`.

Options :

6406534214616. ✖ {x = 5, y = 5, z = 5}, {x = 4, y = 4, z = 5}

6406534214617. ✖ {x = 5, y = 4, z = 5}, {x = 4, y = 4, z = 5}, {x = 4, y = 4, z = 4}

6406534214618. ✖ {x = 5, y = 5, z = 5}, {x = 4, y = 4, z = 5}, {x = 4, y = 4, z = 4}

6406534214619. ✔ {x = 5, y = 4, z = 4}, {x = 5, y = 4, z = 5}, {x = 4, y = 4, z = 4}

Question Number : 211 Question Id : 6406531250878 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Consider the prime path P = [1, 2, 4, 5]. Among the options given, identify the path that tours P with a sidetrip but not a detour.

Options :

6406534214620. ✖ [1, 2, 4, 2, 4, 5]

6406534214621. ✖ [1, 2, 4, 2, 3, 4]

6406534214622. ✔ [1, 2, 3, 2, 4, 5]

6406534214623. ✖ [1, 2, 3, 4, 5]

Question Number : 212 Question Id : 6406531250880 Question Type : MCQ
Correct Marks : 5
Question Label : Multiple Choice Question

Consider the following Java program.

```
class Device {
    protected String name, type;
    protected int memorySizeGB;
    public void setName(String n) { name = n; }
}
class Computer extends Device {
    public Computer() { type = "Computer"; }
    public void setMemorySize(int ms) { this.memorySizeGB = ms; }
    public void printSpecs() {
        System.out.println(name + ", " + type + ", " + memorySizeGB);
    }
}
class Laptop extends Computer {
    protected int memorySizeGB;
    public Laptop() { type = "Laptop"; }
    public void setMemorySize(int ms) { memorySizeGB = ms; }
}
class Process {
    public static void main(String[] args) {
        Laptop laptop = new Laptop();
        laptop.setName("Orange");
        laptop.setMemorySize(16);
        laptop.printSpecs();
    }
}
```

The above program generates output as `Orange, Laptop, 0`, instead of `Orange, Laptop, 16`. The reason is that `Device::memorySizeGB` is overridden in the descendant `Laptop` which hides the inherited variable `Device::memorySizeGB` in class `Laptop`. Identify the type of anomaly or fault in the given scenario.

Options :

6406534214628. ✖ Inconsistent type use

6406534214629. ✖ State definition anomaly

6406534214630. ✔ State definition inconsistency anomaly

6406534214631. ✖ State visibility anomaly

Question Number : 213 Question Id : 6406531250881 Question Type : MCQ
Correct Marks : 5
Question Label : Multiple Choice Question

Consider the following code segment.

```
1.  class Shape {
2.      public void draw() {
3.          System.out.println("Drawing shape");
4.      }
5.  }
6.  class Circle extends Shape {
7.      public void draw() {
8.          super.draw();
9.          System.out.println("Drawing circle");
10.     }
11. }
```

Suppose a mutant of the code is created by deleting LINE-8. What type of mutation operator is applied in this case?

Options :

6406534214632. ✖ Overriding method deletion (OMD)

6406534214633. ✖ Overriding method moving (OMM)

6406534214634. ✖ Overloading method deletion (OMD)

6406534214635. ✔ Super keyword deletion (SKD)

Question Number : 214 Question Id : 6406531250882 Question Type : MCQ
Correct Marks : 5

Question Label : Multiple Choice Question

Consider the following code segment.

```
// Code base
int add(int n){
    return n + 5;
}
int subtract(int n) {
    return n - 3;
}
int multiply(int n) {
    return n * 2;
}
int power(int n) {
    return n * n;
}
int process(int a, int b) {
    int t1 = add(a);
    int t2 = subtract(b);
    if(power(t1) < multiply(t2)) {
        if(t1 > t2) {      // LINE-1

            exit(0);       // LINE-2: Error
        }
        // some other code
    }
    return t2;
}
```

Consider the initial symbolic states as: $a = a_0, b = b_0$.
At LINE-1 two instances of symbolic execution are created. Identify the path constraints applicable to those?

Options :

6406534214636. ✖ $((a_0 + 5) \times (a_0 + 5)) < (2 \times (b_0 - 3))$ and $((a_0 + 5) \times (a_0 + 5)) \geq (2 \times (b_0 - 3))$

6406534214637. ✖ $((a_0 + 5) > (b_0 - 3))$ and $((a_0 + 5) \leq (b_0 - 3))$

6406534214638. ✖ $((a_0 + 5) \times (a_0 + 5)) < (2 \times (b_0 - 3)) \wedge ((a_0 + 5) > (b_0 - 3))$ and $((a_0 + 5) \times (a_0 + 5)) \geq (2 \times (b_0 - 3)) \wedge ((a_0 + 5) > (b_0 - 3))$

6406534214639. ✔ $((a_0 + 5) \times (a_0 + 5)) < (2 \times (b_0 - 3)) \wedge ((a_0 + 5) > (b_0 - 3))$ and $((a_0 + 5) \times (a_0 + 5)) < (2 \times (b_0 - 3)) \wedge ((a_0 + 5) \leq (b_0 - 3))$

Sub-Section Number :	4
Sub-Section Id :	640653194245
Question Shuffling Allowed :	Yes

Question Number : 215 Question Id : 6406531250883 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question
Consider the following partition for a job application portal:

$$58 \leq candidate_weight \leq 78$$

Identify all the valid set(s) of test cases for *candidate_age* (*candidate_age* is considered in years, as a positive integer value only) using Boundary Value Analysis (BVA) for the given partitions.

Options :
6406534214640. ✖ {58, 67, 78}
6406534214641. ✖ {58, 59, 68, 77, 78}
6406534214642. ✔ {57, 58, 67, 78, 79}
6406534214643. ✔ {57, 58, 59, 65, 77, 78, 79}

Sub-Section Number :	5
Sub-Section Id :	640653194246
Question Shuffling Allowed :	No

Question Id : 6406531250884 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (216 to 218)
Question Label : Comprehension
Consider the following Java method below with a mutant indicated in the comments and answer the given subquestions.

```
public static void isPalindrome(String s) {
    String reversed = "";
    // Mutant-1: for (int i = s.length() - 1; i > 0; i--)
    for (int i = s.length() - 1; i >= 0; i--) {
        reversed = reversed + s.charAt(i);
    }
    if (s.equals(reversed)) {
        System.out.println(s + " is a palindrome");
    } else {
        System.out.println(s + " is not a palindrome");
    }
}
```

Sub questions

Question Number : 216 Question Id : 6406531250885 Question Type : MCQ
Correct Marks : 4
Question Label : Multiple Choice Question

State true or false: The mutant Mutant-1 will always cause an infection.

Options :

6406534214644. ✔ TRUE

6406534214645. ✖ FALSE

Question Number : 217 Question Id : 6406531250886 Question Type : MSQ
Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following inputs will not cause propagation for Mutant-1?

Options :

6406534214646. ✖ refer

6406534214647. ✔ river

6406534214648. ✖ level

6406534214649. ✔ little

Question Number : 218 Question Id : 6406531250887 Question Type : MSQ
Correct Marks : 5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following inputs would strongly kill the mutant Mutant-1?

Options :

6406534214650. ✔ refer

6406534214651. ✖ river

6406534214652. ✔ level

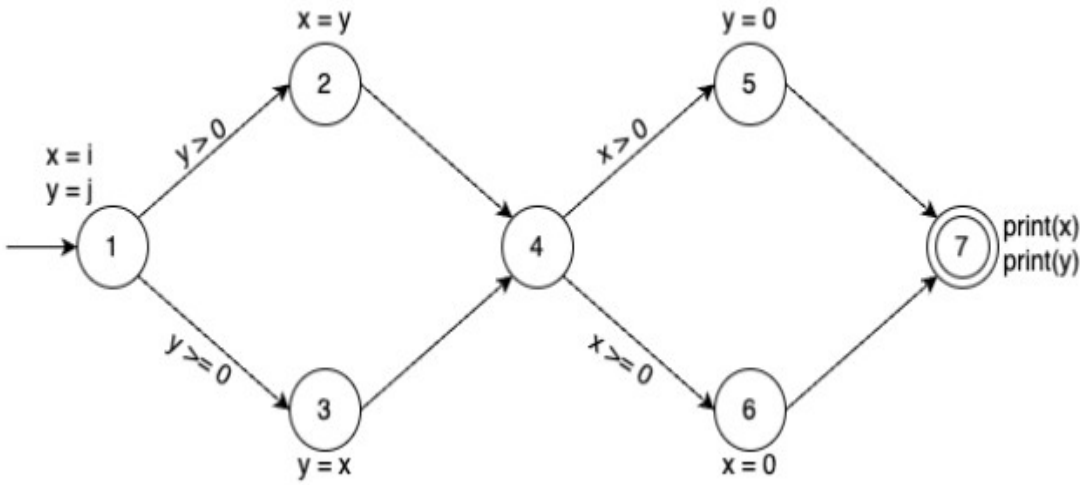
6406534214653. ✖ little

Sub-Section Number :	6
Sub-Section Id :	640653194247
Question Shuffling Allowed :	No

Question Id : 6406531250888 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (219 to 220)

Question Label : Comprehension

Consider the following annotated control flow graph (CFG) and answer the given subquestions.



Sub questions

Question Number : 219 Question Id : 6406531250889 Question Type : MCQ
Correct Marks : 4
Question Label : Multiple Choice Question
Which of the following is a def-clear path with respect to variable **y**?
Options :
6406534214654. ✖ [1, 2, 4, 5, 7]
6406534214655. ✔ [1, 2, 4, 6, 7]
6406534214656. ✖ [1, 3, 4, 5, 7]
6406534214657. ✖ [1, 3, 4, 6, 7]

Question Number : 220 Question Id : 6406531250890 Question Type : MCQ
Correct Marks : 5
Question Label : Multiple Choice Question
Which of the following statements is true about the set of testpaths {[1, 2, 4, 6, 7], [1, 2, 4, 5, 7]} ?
Options :
6406534214658. ✖ It satisfies all defs coverage for variable **y**.
6406534214659. ✖ It satisfies both the all defs coverage, and all uses coverage for variable **y**.
6406534214660. ✖ It satisfies both the all defs coverage, all-use coverage, and all du-paths coverage for variable **y**.
6406534214661. ✔ None of these

Sub-Section Number :	7
Sub-Section Id :	640653194248
Question Shuffling Allowed :	No

Question Id : 6406531250891 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (221 to 222)
Question Label : Comprehension
Consider the predicate $p = a \wedge (\neg b \vee \neg c)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 221 Question Id : 6406531250892 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Which of the following represents p_a, the conditions under which the clause a determines p?

Options :

6406534214662. ✖ $\neg b \vee c$

6406534214663. ✔ $\neg b \vee \neg c$

6406534214664. ✖ $\neg b \wedge \neg c$

6406534214665. ✖ $b \wedge c$

Question Number : 222 Question Id : 6406531250893 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

From the given options, identify all pairs of test requirements to satisfy the restricted inactive clause coverage (RICC) for clause *a*.

Options :

6406534214666. ✖ For $p = T$, $\{(a = F, b = T, c = T), (a = F, b = F, c = T)\}$
No feasible pair for $p = F$,

6406534214667. ✔ No feasible pair for $p = T$,
For $p = F$, $\{(a = T, b = T, c = T),$
$(a = F, b = T, c = T)\}$

6406534214668. ✖ For $p = T$, $\{(a = T, b = T, c = F), (a = F, b = T, c = F)\}$
For $p = F$, $\{(a = T, b = F, c = T), (a = F, b = F, c = T)\}$

6406534214669. ✖ For $p = T$, $\{(a = T, b = T, c = F), (a = F, b = T, c = F)\}$
For $p = F$, $\{(a = T, b = T, c = F), (a = F, b = F, c = T)\}$,
$\{(a = T, b = F, c = T), (a = F, b = T, c = F)\}$